

PAIC simulation results

2023-10-04

Overview of the problem

The goal is to estimate the relative effect of treatment A vs B , noted Δ_t . These two treatments have been evaluated in two distinct trials, AB and AC . The target estimand is the relative effect of A vs B in the source population of trial AC , written $\Delta_{t(AC)}$.

$\Delta_{t(AB)}$ will be estimated using different estimators:

- Unadjusted:
 - Unanchored: naïve model
 - Anchored: difference of relative effects A versus C and B versus C
- Adjusted
 - Adjusted using a multivariable outcome model (regression model), or univariable weighted outcome model (propensity score)
 - * Each of these models will be fitted using either Individual Patient Data (IPD) for both trials, or IPD for the trial AB and AgD for trial BC : Population-Adjusted Indirect Comparisons (PAIC). PAIC multivariable outcome model = Simulated Treatment Comparison (STC), univariable weighted outcome model = Matching-Adjusted Indirect Comparisons (MAIC).
 - * Each of these models will be either anchored or unanchored.

(Current) scenario settings

- Sample size per treatment arm = 100
- One gaussian outcome Y
- Treatment effects:
 - $\beta_A = 1.5$
 - $\beta_B = 1$
 - $\beta_C = 0$ (*no effect, ~ placebo*)
- One binary confounding factor X_1 , following a binomial distribution $B(X_1) \sim 0.5$ in an overarching population from which patients of trials AB and AC have been sampled. Effects of this confounding factor on 1. trial assignment and 2. the outcome Y :
 1. Constant effect on the outcome (prognostic): $\beta_{Y_{X_1}} = 1.5$, and effect on the outcome through interaction with treatment A (interaction): $\beta_{Y_{A:X_1}} = 1.2$
 2. Effect on trial assignment ($\beta_{AC,X_1} = 2$) and ($\beta_{BC,X_1} = 0$)
- Outcome model :

$$Y \sim (\beta_{Y_{X_1:A}}A + \beta_{Y_{X_1}})X_1 + \beta_{Y_A}A + \beta_{Y_B}B + \beta_{Y_C}C + \epsilon_Y$$

with $\epsilon_Y \sim N(0, 1)$

- Trial assignment model : $P(Z = AC) \sim g(\beta_{AC,X_1}X_1)$ with $g(X) = (1 + \exp(-X))^{-1}$

```
prop_X1 <- 0.5
bT_X1 <- 2

bY_X1 <- 1.5
bYA_X1 <- 1.2

bY_A <- 1.5
bY_B <- 1.5
bY_C <- 0

N_RCT <- 200

trial_assignment_model_AC <- bquote(X1 * bT_X1)
trial_assignment_model_BC <- bquote(0)
outcome_model <- bquote(bYA_X1*X1*A + bY_X1*X1 + bY_A*A + bY_B*B + bY_C*C)
```

Monte-Carlo simulations settings

Two thousands iteration of that scenario, and estimation of variance of treatment effects via bootstrap (1000 iterations).

```
library(data.table)
N_pop <- 10^6

pop_init <- data.table(
  id = 1:N_pop,
  X1 = rbinom(N_pop, 1, prop_X1)
)

trial_assignment_prob <- function(df, trial_assignment_model, deviates = FALSE) {
  predicted <- with(df, eval(trial_assignment_model))
  # variance of a logistic function is set ((s^2\pi^2)/3), with s=1 when using normal law -->
  # making an assumption about bias and residual heterogeneity
  return(plogis(predicted))
}

predict_outcome <- function(df, outcome_model, deviates = FALSE) {
  predicted <- with(df, eval(outcome_model))
  if (deviates) return(predicted + rnorm(n = length(predicted), mean = 0, sd = 1)) else return(predicted)
}

pop_init$prob_AC <- trial_assignment_prob(pop_init, trial_assignment_model_AC)
pop_init$prob_BC <- trial_assignment_prob(pop_init, trial_assignment_model_BC)

pop_init$Y_theo_A <- pop_init[, c("A", "B", "C") := .(1L, 0L, 0L)] |>
  predict_outcome(outcome_model)
pop_init$Y_theo_B <- pop_init[, c("A", "B", "C") := .(0L, 1L, 0L)] |>
```

```

  predict_outcome(outcome_model)
pop_init$Y_theo_C <- pop_init[, c("A", "B", "C")] := .(OL, OL, 1L)] |>
  predict_outcome(outcome_model)
pop_init[, c("A", "B", "C")] := NULL]

pop_init$prob_AC <- trial_assignment_prob(pop_init, trial_assignment_model_AC)
pop_init$prob_BC <- trial_assignment_prob(pop_init, trial_assignment_model_BC)

ttn_names <- c("A", "B", "C")
cols_obs <- paste0("Y_obs_", ttn_names)
cols_theo <- paste0("Y_theo_", ttn_names)
# Assigning "observed" values for all the individual in the population, ie theoretical outcome + gaussian noise
pop_init[, eval(bquote(cols_obs)) := lapply(.SD, \(x) x + rnorm(n = length(x))), .SD = cols_theo]

```

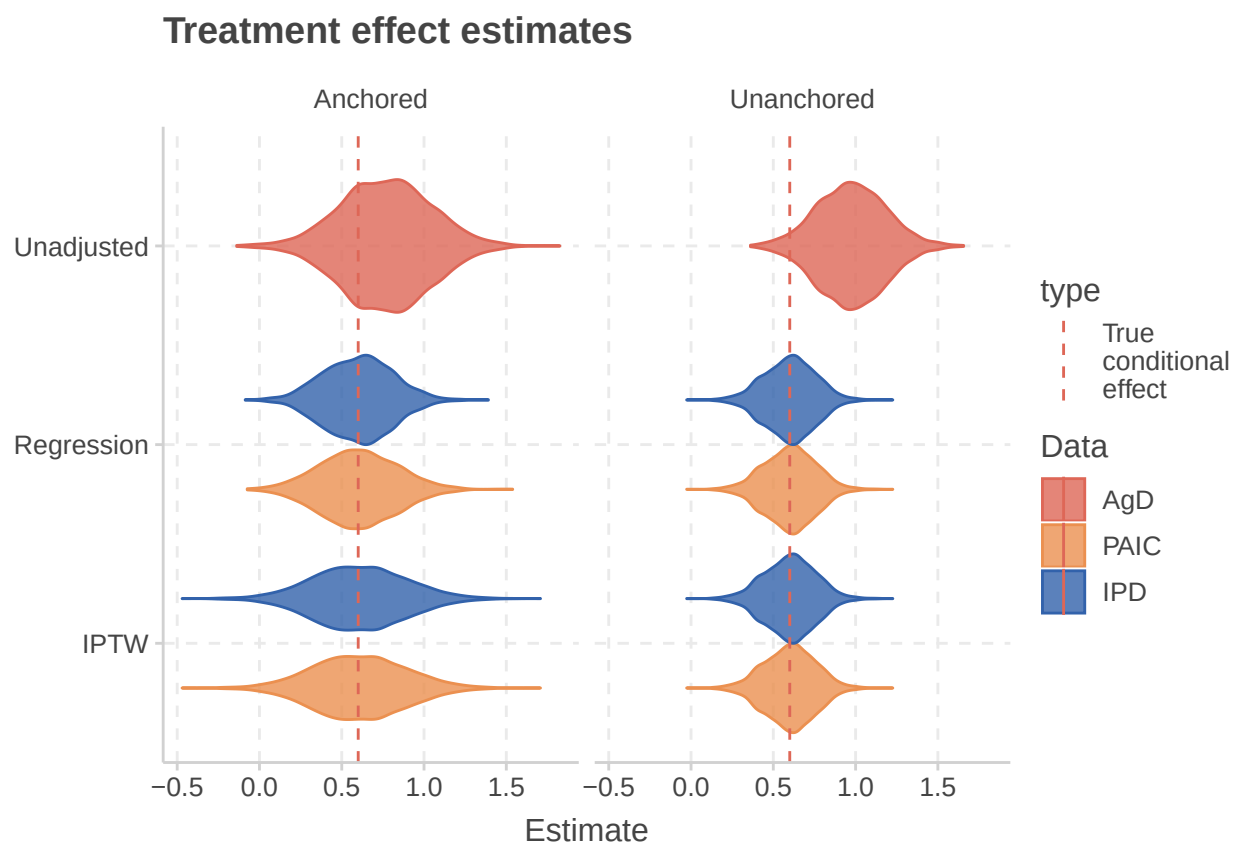
```

trial_AC <- pop_init[
  sample(pop_init$id, N_RCT, replace = FALSE, prob = pop_init$prob_AC),
  .(id, ttn = factor(rep_len(c("A", "C"), length.out = N_RCT), levels = c("C", "A", "B")))] [
  pop_init_long[,c("id", "X1", "ttn", "prob_AC", "theo", "obs")], on = .(id, ttn), nomatch = NULL
]
trial_BC <- pop_init[
  sample(pop_init$id, N_RCT, replace = FALSE, prob = pop_init$prob_BC),
  .(id, ttn = factor(rep_len(c("B", "C"), length.out = N_RCT), levels = c("C", "A", "B")))] [
  pop_init_long[,c("id", "X1", "ttn", "prob_BC", "theo", "obs")], on = .(id, ttn), nomatch = NULL
]

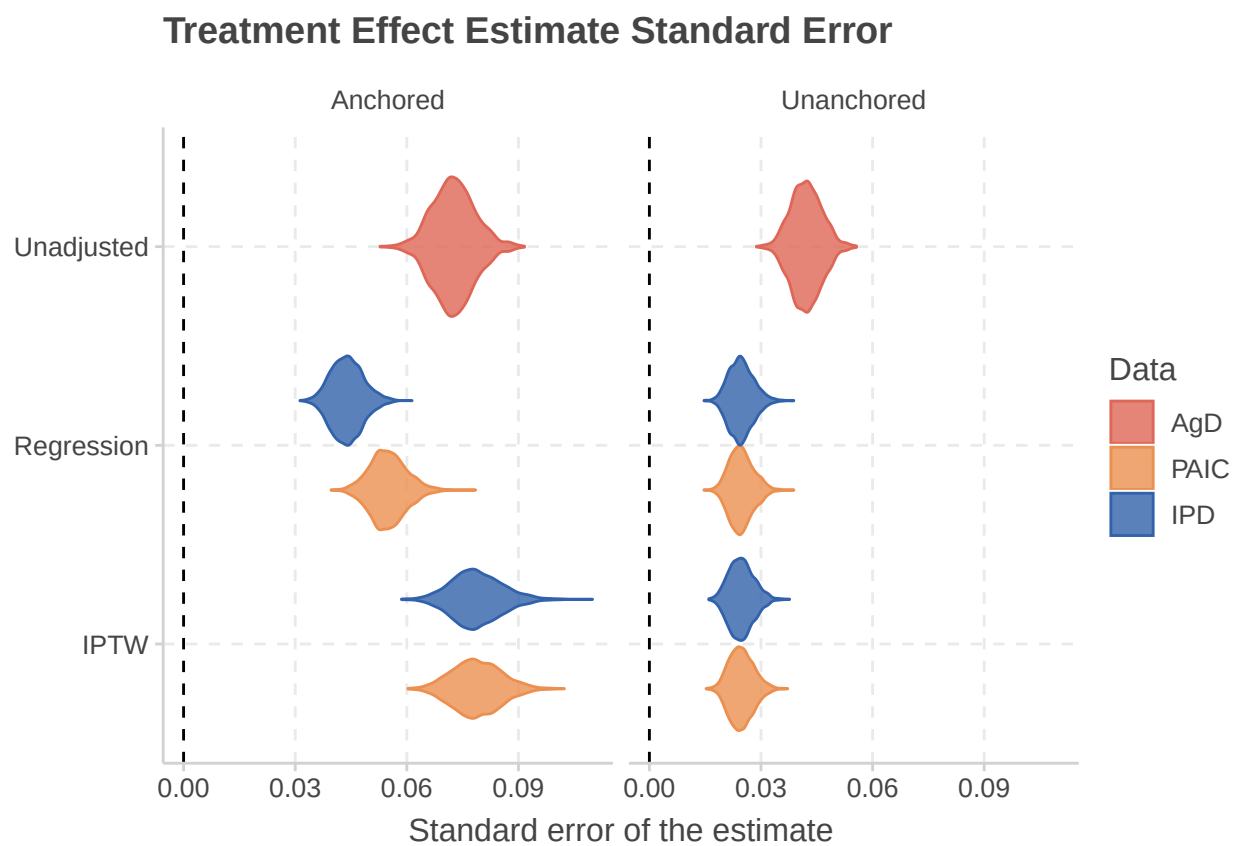
```

Simulation results

Distribution of the estimate



Standard error of the estimate



Performance of the estimators

Performance of the estimators

